

1. Create Table on the basis of table and data given above. Data types and size have to be decided by you and may vary.

The screenshot shows the MySQL Workbench interface. The main editor displays the following SQL queries:

```
1 create DATABASE Saad;
2 use Saad;
3
4 /*1. Create Table on the basis of table and data given above.
5 Data types and size have to be decided by you and may vary.*/
6
7 CREATE TABLE Customers_Table (
8 CustomerID INT PRIMARY KEY,
9 FirstName VARCHAR(30),
10 LastName VARCHAR(30),
11 Age INT,
12 Gender VARCHAR(1),
13 Address VARCHAR(60),
14 City VARCHAR(40),
15 State VARCHAR(10),
16 ZipCode INT,
17 Email VARCHAR(40)
18 );
19
20 SELECT
21 *
22 FROM
23 Customers_Table;
```

The output pane shows the results of the queries:

#	Time	Action	Message	Duration / Fetch
5	16:24:32	CREATE TABLE Customers_Table(CustomerID Int PRIMARY KEY, FirstNa...	Error Code: 1050. Table 'customers_table' already exists	0.000 sec
6	16:24:49	SELECT * FROM Customers_Table LIMIT 0, 1000	0 row(s) returned	0.000 sec / 0.000 sec
7	16:25:58	DELETE Customers_Table	Error Code: 1064. You have an error in your SQL syntax; check the manual ...	0.016 sec
8	16:26:05	SELECT * FROM Customers_Table LIMIT 0, 1000	0 row(s) returned	0.000 sec / 0.000 sec

2. Also Insert data into table as described above.

MySQL Workbench

Local instance MySQL80

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Administration Schemas

Information

No object selected

Query 1

Limit to 1000 rows

```

23
24 /*2. Also Insert data into table as described above.*/
25
26 INSERT INTO Customers_Table (CustomerID, FirstName, LastName, Age, Gender, Address, City, State, ZipCode, Email) VALUES
27 (1, 'Alice', 'Johnson', 30, 'F', '123 Elm St', 'New York', 'NY', 10001, 'alicejohnson@example.com'),
28 (2, 'Bob', 'Williams', 45, 'M', '456 Maple St', 'Los Angeles', 'CA', 90001, 'bobwilliams@example.com'),
29 (3, 'Charlie', 'Brown', 49, 'M', '789 Oak St', 'Chicago', 'IL', 60601, 'charliebrown@example.com'),
30 (4, 'Diana', 'Evans', 34, 'F', '321 Pine St', 'Houston', 'TX', 77001, 'dianaevans@example.com'),
31 (5, 'Ethan', 'Wright', 37, 'M', '654 Birch St', 'Phoenix', 'AZ', 85001, 'ethanwright@example.com'),
32 (6, 'Fiona', 'Adams', 26, 'F', '987 Cedar St', 'Philadelphia', 'PA', 19101, 'fionaadams@example.com'),
33 (7, 'George', 'Miller', 33, 'M', '111 Spruce St', 'San Diego', 'CA', 92101, 'georgemiller@example.com'),
34 (8, 'Hannah', 'Taylor', 30, 'F', '222 Willow St', 'Dallas', 'TX', 75201, 'hannahtaylor@example.com'),
35 (9, 'Ian', 'Thomas', 25, 'M', '333 Cherry St', 'San Jose', 'CA', 95101, 'ianthomas@example.com'),
36 (10, 'Julia', 'Wilson', 36, 'F', '444 Poplar St', 'Austin', 'TX', 73301, 'juliawilson@example.com'),
37 (11, 'Kevin', 'Lee', 42, 'M', '555 Aspen St', 'Denver', 'CO', 80201, 'kevinlee@example.com'),
38 (12, 'Laura', 'Hall', 28, 'F', '666 Hickory St', 'Seattle', 'WA', 98101, 'laurahall@example.com'),
39 (13, 'Mark', 'Harris', 31, 'M', '777 Redwood St', 'Portland', 'OR', 97201, 'markharris@example.com'),
40 (14, 'Natalie', 'King', 27, 'F', '888 Cypress St', 'Miami', 'FL', 33101, 'natalieking@example.com'),
41 (15, 'Oliver', 'Lewis', 44, 'M', '999 Chestnut St', 'Boston', 'MA', 02101, 'oliverlewis@example.com'),
42 (16, 'Peter', 'Carter', 41, 'M', '101 Walnut St', 'Atlanta', 'GA', 30301, 'petercarter@example.com'),
43 (17, 'Rachel', 'Adams', 29, 'F', '202 Pineapple St', 'Nashville', 'TN', 37201, 'racheladams@example.com'),
44 (18, 'Samuel', 'Clark', 39, 'M', '303 Magnolia St', 'New Orleans', 'LA', 70101, 'samuelclark@example.com'),
45 (19, 'Teresa', 'Scott', 40, 'F', '404 Peach St', 'Charlotte', 'NC', 28201, 'teresascott@example.com'),
46 (20, 'Victor', 'Young', 37, 'M', '505 Palm St', 'Tampa', 'FL', 33601, 'victoryoung@example.com'),
47 (21, 'Wendell', 'Decker', 46, 'M', '606 Diamond St', 'Indianapolis', 'IN', 46201, 'wendelldecker@example.com')

```

Result Grid

CustomerID	FirstName	LastName	Age	Gender	Address	City	State	ZipCode	Email
1	Alice	Johnson	30	F	123 Elm St	New York	NY	10001	alicejohnson@example.com
2	Bob	Williams	45	M	456 Maple St	Los Angeles	CA	90001	bobwilliams@example.com
3	Charlie	Brown	49	M	789 Oak St	Chicago	IL	60601	charliebrown@example.com
4	Diana	Evans	34	F	321 Pine St	Houston	TX	77001	dianaevans@example.com
5	Ethan	Wright	37	M	654 Birch St	Phoenix	AZ	85001	ethanwright@example.com
6	Fiona	Adams	26	F	987 Cedar St	Philadelphia	PA	19101	fionaadams@example.com
7	George	Miller	33	M	111 Spruce St	San Diego	CA	92101	georgemiller@example.com

Customers_Table 4 x

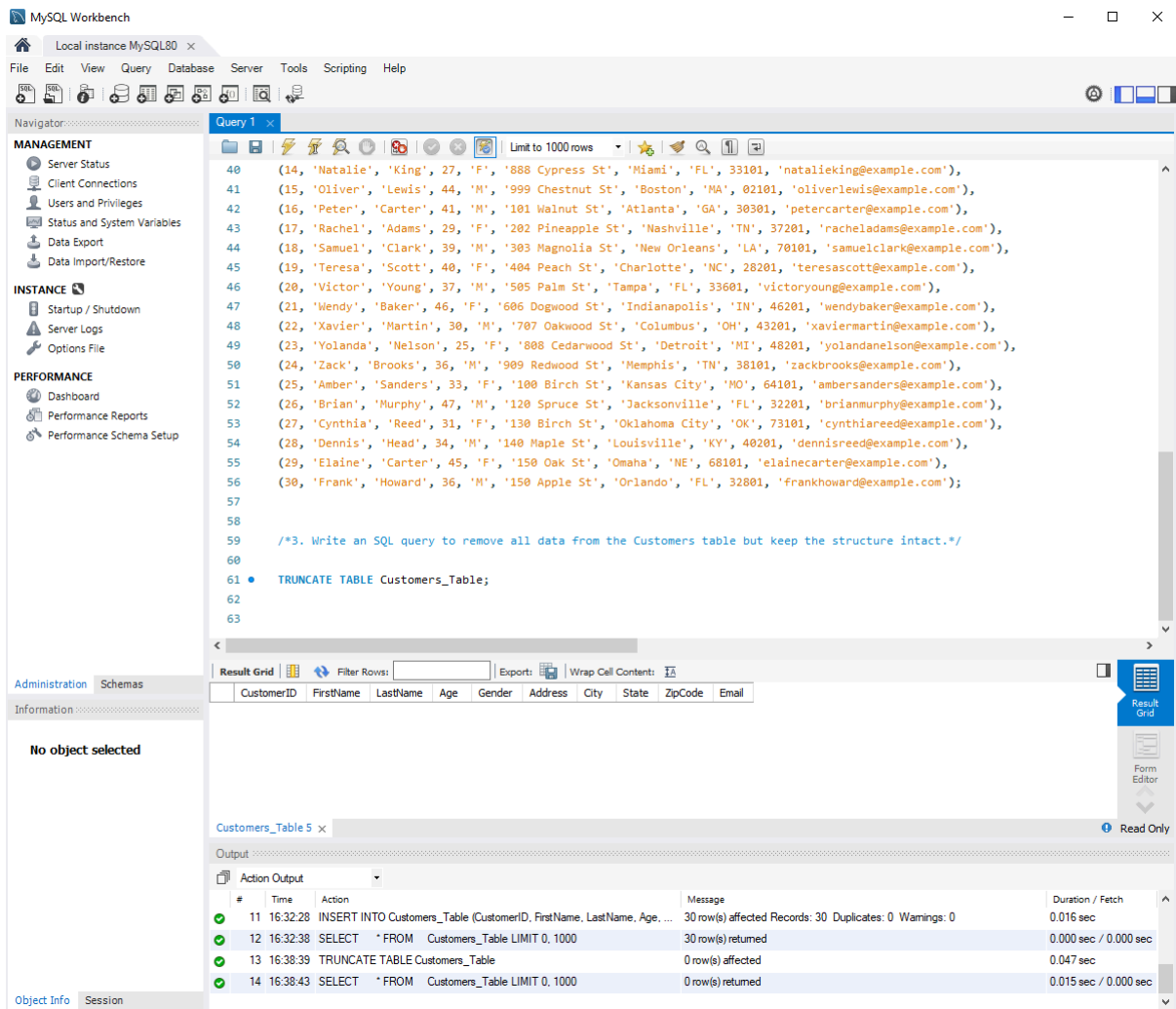
Output

Action Output

#	Time	Action	Message	Duration / Fetch
9	16:30:42	INSERT INTO users (id, first_name, last_name, age, gender, address, city, ...	Error Code: 1146. Table 'saad.users' doesn't exist	0.000 sec
10	16:32:14	INSERT INTO users (CustomerID, FirstName, LastName, Age, Gender, Add...	Error Code: 1146. Table 'saad.users' doesn't exist	0.000 sec
11	16:32:28	INSERT INTO Customers_Table (CustomerID, FirstName, LastName, Age, ...	30 row(s) affected Records: 30 Duplicates: 0 Warnings: 0	0.016 sec
12	16:32:38	SELECT * FROM Customers_Table LIMIT 0, 1000	30 row(s) returned	0.000 sec / 0.000 sec

Object Info Session

3. Write an SQL query to remove all data from the Customers table but keep the structure intact.



4. What is the key difference between TRUNCATE and DELETE?

Ans. The DROP TABLE command deletes a table in the database. The TRUNCATE TABLE command deletes the data inside a table, but not the table itself.

5. Write an SQL query to find customers aged between 25 and 45.

MySQL Workbench

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Query 1

```

47 (21, 'Wendy', 'Baker', 46, 'F', '606 Dogwood St', 'Indianapolis', 'IN', 46201, 'wendybaker@example.com'),
48 (22, 'Xavier', 'Martin', 30, 'M', '707 Oakwood St', 'Columbus', 'OH', 43201, 'xaviermartin@example.com'),
49 (23, 'Yolanda', 'Nelson', 25, 'F', '808 Cedarwood St', 'Detroit', 'MI', 48201, 'yolandanelson@example.com'),
50 (24, 'Zack', 'Brooks', 36, 'M', '909 Redwood St', 'Memphis', 'TN', 38101, 'zackbrooks@example.com'),
51 (25, 'Amber', 'Sanders', 33, 'F', '100 Birch St', 'Kansas City', 'MO', 64101, 'ambersanders@example.com'),
52 (26, 'Brian', 'Murphy', 47, 'M', '120 Spruce St', 'Jacksonville', 'FL', 32201, 'brianmurphy@example.com'),
53 (27, 'Cynthia', 'Reed', 31, 'F', '130 Birch St', 'Oklahoma City', 'OK', 73101, 'cynthiareed@example.com'),
54 (28, 'Dennis', 'Head', 34, 'M', '140 Maple St', 'Louisville', 'KY', 40201, 'dennisreed@example.com'),
55 (29, 'Elaine', 'Carter', 45, 'F', '150 Oak St', 'Omaha', 'NE', 68101, 'elainecarter@example.com'),
56 (30, 'Frank', 'Howard', 36, 'M', '150 Apple St', 'Orlando', 'FL', 32801, 'frankhoward@example.com');
57
58
59 /*3. Write an SQL query to remove all data from the Customers table but keep the structure intact.*/
60
61 TRUNCATE TABLE Customers_Table;
62
63 /*5. Write an SQL query to find customers aged between 25 and 45.*/
64
65 SELECT
66     CustomerID, FirstName, LastName, Age
67 FROM
68     Customers_Table
69 WHERE
70     Age BETWEEN 25 AND 45;

```

Result Grid

CustomerID	FirstName	LastName	Age
1	Alice	Johnson	30
2	Bob	Williams	45
4	Diana	Evans	34
5	Ethan	Wright	37
6	Fiona	Adams	26
7	George	Miller	33
8	Hannah	Taylor	30

Customers_Table 7

Output

#	Time	Action	Message	Duration / Fetch
14	16:38:43	SELECT * FROM Customers_Table LIMIT 0, 1000	0 row(s) returned	0.015 sec / 0.000 sec
15	16:39:45	INSERT INTO Customers_Table (CustomerID, FirstName, LastName, Age, ...	30 row(s) affected Records: 30 Duplicates: 0 Warnings: 0	0.000 sec
16	16:39:50	SELECT * FROM Customers_Table LIMIT 0, 1000	30 row(s) returned	0.000 sec / 0.000 sec
17	16:42:26	SELECT CustomerID, FirstName, LastName, Age FROM Customers_T...	27 row(s) returned	0.016 sec / 0.000 sec

Object Info Session

6. Retrieve customers whose Zip Code falls between '10000' and '50000'.

MySQL Workbench

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Query 1

```

56 (30, 'Frank', 'Howard', 36, 'M', '150 Apple St', 'Orlando', 'FL', 32801, 'Frankhoward@example.com');
57
58
59 /*3. Write an SQL query to remove all data from the Customers table but keep the structure intact.*/
60
61 • TRUNCATE TABLE Customers_Table;
62
63 /*5. Write an SQL query to find customers aged between 25 and 45.*/
64
65 • SELECT
66     CustomerID, FirstName, LastName, Age
67 FROM
68     Customers_Table
69 WHERE
70     Age BETWEEN 25 AND 45;
71
72 /*6. Retrieve customers whose Zip Code falls between 10000 and 50000.*/
73
74 • SELECT
75     CustomerID, FirstName, LastName, ZipCode
76 FROM
77     Customers_Table
78 WHERE
79     ZipCode BETWEEN 10000 AND 50000;

```

Result Grid

CustomerID	FirstName	LastName	ZipCode
1	Alice	Johnson	10001
6	Fiona	Adams	19101
14	Natalie	King	33101
16	Peter	Carter	30301
17	Rachel	Adams	37201
19	Teresa	Scott	28201
20	Victor	Young	33601

Customers_Table 8

Output

#	Time	Action	Message	Duration / Fetch
✓ 15	16:39:45	INSERT INTO Customers_Table (CustomerID, FirstName, LastName, Age, ...	30 row(s) affected Records: 30 Duplicates: 0 Warnings: 0	0.000 sec
✓ 16	16:39:50	SELECT * FROM Customers_Table LIMIT 0, 1000	30 row(s) returned	0.000 sec / 0.000 sec
✓ 17	16:42:26	SELECT CustomerID, FirstName, LastName, Age FROM Customers_T...	27 row(s) returned	0.016 sec / 0.000 sec
✓ 18	16:44:42	SELECT CustomerID, FirstName, LastName, ZipCode FROM Custome...	14 row(s) returned	0.000 sec / 0.000 sec

Object Info Session

7. Get a list of customers with Customer Id between 10 and 20, ordered by LastName.

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Information

No object selected

Query 1

```

66 CustomerID, FirstName, LastName, Age
67 FROM
68 Customers_Table
69 WHERE
70 Age BETWEEN 25 AND 45;
71
72 /*6. Retrieve customers whose Zip Code falls between 10000 and 50000;*/
73
74 • SELECT
75 CustomerID, FirstName, LastName, ZipCode
76 FROM
77 Customers_Table
78 WHERE
79 ZipCode BETWEEN 10000 AND 50000;
80
81 /*7. Get a list of customers with Customer Id between 10 and 20, ordered by LastName.*/
82
83 • SELECT
84 CustomerID, FirstName, LastName
85 FROM
86 Customers_Table
87 WHERE
88 CustomerID BETWEEN 10 AND 20
89 ORDER BY LastName;

```

Result Grid

CustomerID	FirstName	LastName
17	Rachel	Adams
16	Peter	Carter
18	Samuel	Clark
12	Laura	Hall
13	Mark	Harris
14	Natalie	King
11	Kevin	Lee

Output

#	Time	Action	Message	Duration / Fetch
17	16:42:26	SELECT	CustomerID, FirstName, LastName, Age FROM Customers_T...	27 row(s) returned 0.016 sec / 0.000 sec
18	16:44:42	SELECT	CustomerID, FirstName, LastName, ZipCode FROM Custome...	14 row(s) returned 0.000 sec / 0.000 sec
19	16:46:52	SELECT	CustomerID, FirstName, LastName FROM Customers_Table ...	0 row(s) returned 0.000 sec / 0.000 sec
20	16:47:10	SELECT	CustomerID, FirstName, LastName FROM Customers_Table ...	11 row(s) returned 0.000 sec / 0.000 sec

8. Write a query to find customers not in the age range of 30 to 50.

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Query 1

```

75 CustomerID, FirstName, LastName, ZipCode
76 FROM
77 Customers_Table
78 WHERE
79 ZipCode BETWEEN 10000 AND 50000;
80
81 /*7. Get a list of customers with Customer Id between 10 and 20, ordered by LastName.*/
82
83 • SELECT
84 CustomerID, FirstName, LastName
85 FROM
86 Customers_Table
87 WHERE
88 CustomerID BETWEEN 10 AND 20
89 ORDER BY LastName;
90
91 /*8. Write a query to find customers not in the age range of 30 to 50.*/
92
93 • SELECT
94 CustomerID, FirstName, LastName
95 FROM
96 Customers_Table
97 WHERE
98 Age NOT BETWEEN 30 AND 50;

```

Result Grid

CustomerID	FirstName	LastName
6	Fiona	Adams
9	Ian	Thomas
12	Laura	Hall
14	Natalie	King
17	Rachel	Adams
23	Yolanda	Nelson

Customers_Table 11

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓ 18	16:44:42	SELECT	CustomerID, FirstName, LastName, ZipCode FROM Customers_Table ... 14 row(s) returned	0.000 sec / 0.000 sec
✓ 19	16:46:52	SELECT	CustomerID, FirstName, LastName FROM Customers_Table ... 0 row(s) returned	0.000 sec / 0.000 sec
✓ 20	16:47:10	SELECT	CustomerID, FirstName, LastName FROM Customers_Table ... 11 row(s) returned	0.000 sec / 0.000 sec
✓ 21	16:48:41	SELECT	CustomerID, FirstName, LastName FROM Customers_Table ... 6 row(s) returned	0.000 sec / 0.000 sec

Object Info Session

9. Retrieve all customers sorted by FirstName in ascending order.

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Administration Schemas

Information

No object selected

Query 1

```

79      ZipCode BETWEEN 10000 AND 50000;
80
81      /*7. Get a list of customers with Customer Id between 10 and 20, ordered by LastName.*/
82
83      • SELECT
84        *
85      FROM
86        Customers_Table
87      WHERE
88        CustomerID BETWEEN 10 AND 20
89      ORDER BY LastName;
90
91      /*8. Write a query to find customers not in the age range of 30 to 50.*/
92
93      • SELECT
94        *
95      FROM
96        Customers_Table
97      WHERE
98        Age NOT BETWEEN 30 AND 50;
99
100     /*9. Retrieve all customers sorted by FirstName in ascending order.*/
101
102     • SELECT * FROM Customers_Table ORDER BY FirstName ASC;

```

Result Grid

CustomerID	FirstName	LastName	Age	Gender	Address	City	State	ZipCode	Email
1	Alice	Johnson	30	F	123 Elm St	New York	NY	10001	alicejohnson@example.com
25	Amber	Sanders	33	F	100 Birch St	Kansas City	MO	64101	ambersanders@example.com
2	Bob	Williams	45	M	456 Maple St	Los Angeles	CA	90001	bobwilliams@example.com
26	Brian	Murphy	47	M	120 Spruce St	Jacksonville	FL	32201	brianmurphy@example.com
3	Charlie	Brown	49	M	789 Oak St	Chicago	IL	60601	charlebrown@example.com
27	Cynthia	Reed	31	F	130 Birch St	Oklahoma City	OK	73101	cynthiareed@example.com
28	Dennis	Head	34	M	140 Maple St	Louisville	KY	40201	dennisreed@example.com

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Output

Action Output

#	Time	Action	Message	Duration / Fetch
25	16:51:51	SELECT * FROM Customers_Table WHERE ZipCode BETWEEN 10000 AND 50000	14 row(s) returned	0.000 sec / 0.000 sec
26	16:52:12	SELECT * FROM Customers_Table WHERE Age BETWEEN 25 AND 50	27 row(s) returned	0.015 sec / 0.000 sec
27	16:54:19	SELECT * FROM Customers_Table ORDER BY FirstName IN ASC	Error Code: 1064. You have an error in your SQL syntax; check the manual ...	0.000 sec
28	16:54:52	SELECT * FROM Customers_Table ORDER BY FirstName ASC LIMIT 0, 1...	30 row(s) returned	0.000 sec / 0.000 sec

Object Info Session

10. Write a query to list customers sorted by Age in descending order.

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Administration Schemas

Information

No object selected

Query 1

```

83 • SELECT
84 *
85 FROM
86 Customers_Table
87 WHERE
88 CustomerID BETWEEN 10 AND 20
89 ORDER BY LastName;
90
91 /*8. Write a query to find customers not in the age range of 30 to 50.*/
92
93 • SELECT
94 *
95 FROM
96 Customers_Table
97 WHERE
98 Age NOT BETWEEN 30 AND 50;
99
100 /*9. Retrieve all customers sorted by FirstName in ascending order.*/
101
102 • SELECT * FROM Customers_Table ORDER BY FirstName ASC;
103
104 /*10. Write a query to list customers sorted by Age in descending order.*/
105
106 • SELECT * FROM Customers_Table ORDER BY Age DESC;

```

Result Grid

CustomerID	FirstName	LastName	Age	Gender	Address	City	State	ZipCode	Email
3	Charlie	Brown	49	M	789 Oak St	Chicago	IL	60601	charlebrown@example.com
26	Brian	Murphy	47	M	120 Spruce St	Jacksonville	FL	32201	brianmurphy@example.com
21	Wendy	Baker	46	F	606 Dogwood St	Indianapolis	IN	46201	wendybaker@example.com
2	Bob	Williams	45	M	456 Maple St	Los Angeles	CA	90001	bobwilliams@example.com
29	Elaine	Carter	45	F	150 Oak St	Omaha	NE	68101	elainecarter@example.com
15	Oliver	Lewis	44	M	999 Chestnut St	Boston	MA	2101	oliverlewis@example.com
11	Kevin	Lee	42	M	555 Aspen St	Denver	CO	80201	kevinlee@example.com

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Output

Action Output

#	Time	Action	Message	Duration / Fetch
26	16:52:12	SELECT * FROM Customers_Table WHERE Age BETWEEN 25 A...	27 row(s) returned	0.015 sec / 0.000 sec
27	16:54:19	SELECT * FROM Customers_Table ORDER BY FirstName IN ASC	Error Code: 1064. You have an error in your SQL syntax; check the manual ...	0.000 sec
28	16:54:52	SELECT * FROM Customers_Table ORDER BY FirstName ASC LIMIT 0, 1...	30 row(s) returned	0.000 sec / 0.000 sec
29	16:59:57	SELECT * FROM Customers_Table ORDER BY Age DESC LIMIT 0, 1000	30 row(s) returned	0.000 sec / 0.000 sec

Object Info Session

11. Get all customers ordered first by State (ascending) and then by City (descending).

MySQL Workbench

main

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SCHEMAS

Filter objects

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Information

No object selected

Query 1

```

97 WHERE
98 Age NOT BETWEEN 30 AND 50;
99
100 /*9. Retrieve all customers sorted by FirstName in ascending order.*/
101
102 • SELECT
103 *
104 FROM
105 Customers_Table
106 ORDER BY FirstName ASC;
107
108 /*10. Write a query to list customers sorted by Age in descending order.*/
109
110 • SELECT
111 *
112 FROM
113 Customers_Table
114 ORDER BY Age DESC;
115
116 /*11. Get all customers ordered first by State (ascending) and then by City (descending).*/
117
118 • SELECT * FROM Customers_Table ORDER BY State ASC, City DESC;
119

```

Result Grid

CustomerID	FirstName	LastName	Age	Gender	Address	City	State	ZipCode	Email
5	Ethan	Wright	37	M	654 Birch St	Phoenix	AZ	85001	ethanwright@example.com
9	Ian	Thomas	25	M	333 Cherry St	San Jose	CA	95101	ianthomas@example.com
7	George	Miller	33	M	111 Spruce St	San Diego	CA	92101	georgemiller@example.com
2	Bob	Williams	45	M	456 Maple St	Los Angeles	CA	90001	bobwilliams@example.com
11	Kevin	Lee	42	M	555 Aspen St	Denver	CO	80201	kevinlee@example.com
30	Victor	Young	37	M	300 Palm St	Tampa	FL	33601	victoryoung@example.com
30	Frank	Howard	36	M	100 Apple St	Orlando	FL	32801	frankhoward@example.com
14	Natalie	King	27	F	888 Cypress St	Miami	FL	33101	nataleking@example.com

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Output

Action Output

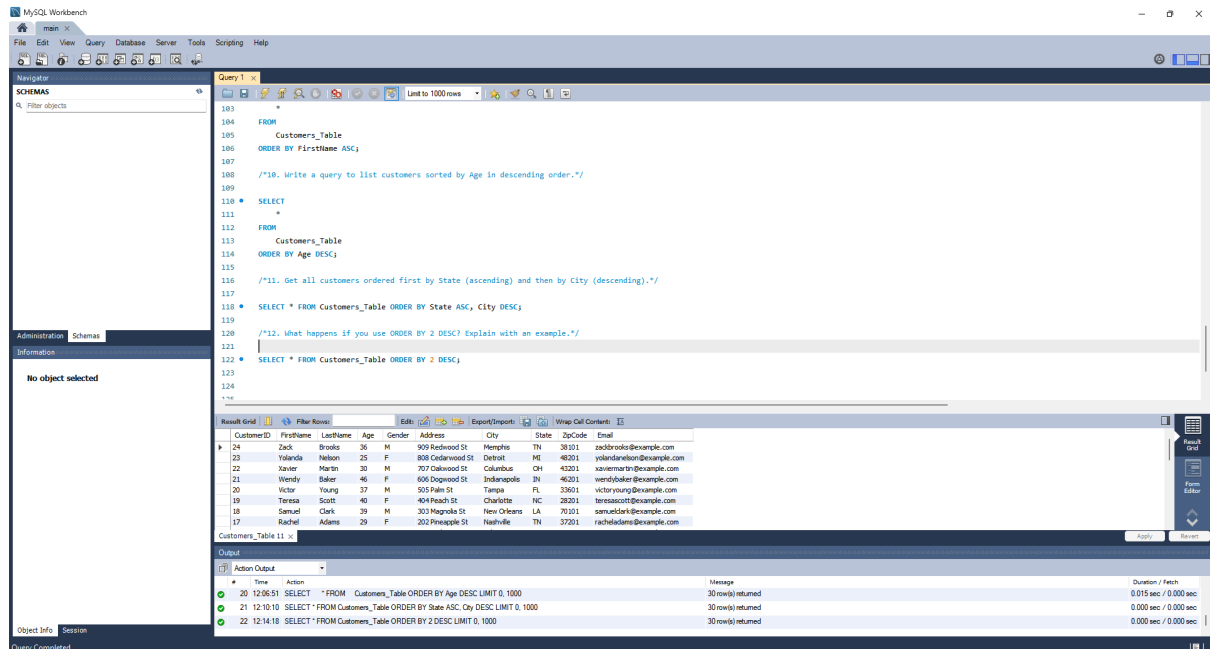
#	Time	Action	Message	Duration / Fetch
19	12:06:47	SELECT * FROM Customers_Table ORDER BY FirstName ASC LIMIT 0, 1000	30 row(s) returned	0.000 sec / 0.000 sec
20	12:06:51	SELECT * FROM Customers_Table ORDER BY Age DESC LIMIT 0, 1000	30 row(s) returned	0.015 sec / 0.000 sec
21	12:10:10	SELECT * FROM Customers_Table ORDER BY State ASC, City DESC LIMIT 0, 1000	30 row(s) returned	0.000 sec / 0.000 sec

Object Info Session

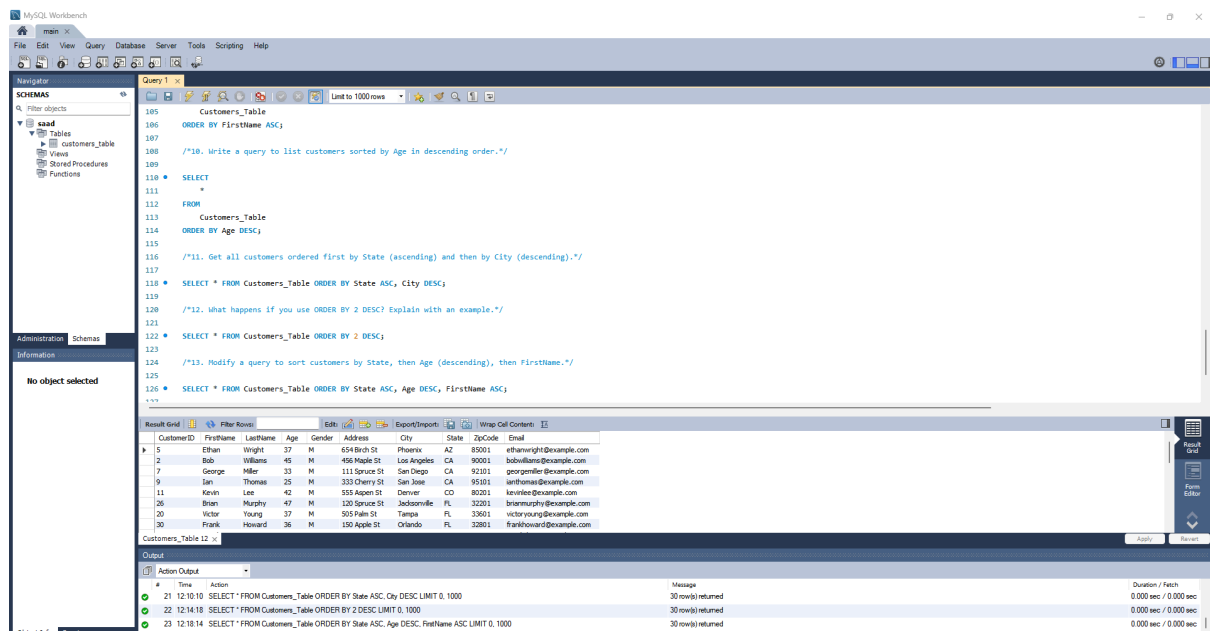
Query Completed

12. What happens if you use ORDER BY 2 DESC? Explain with an example.

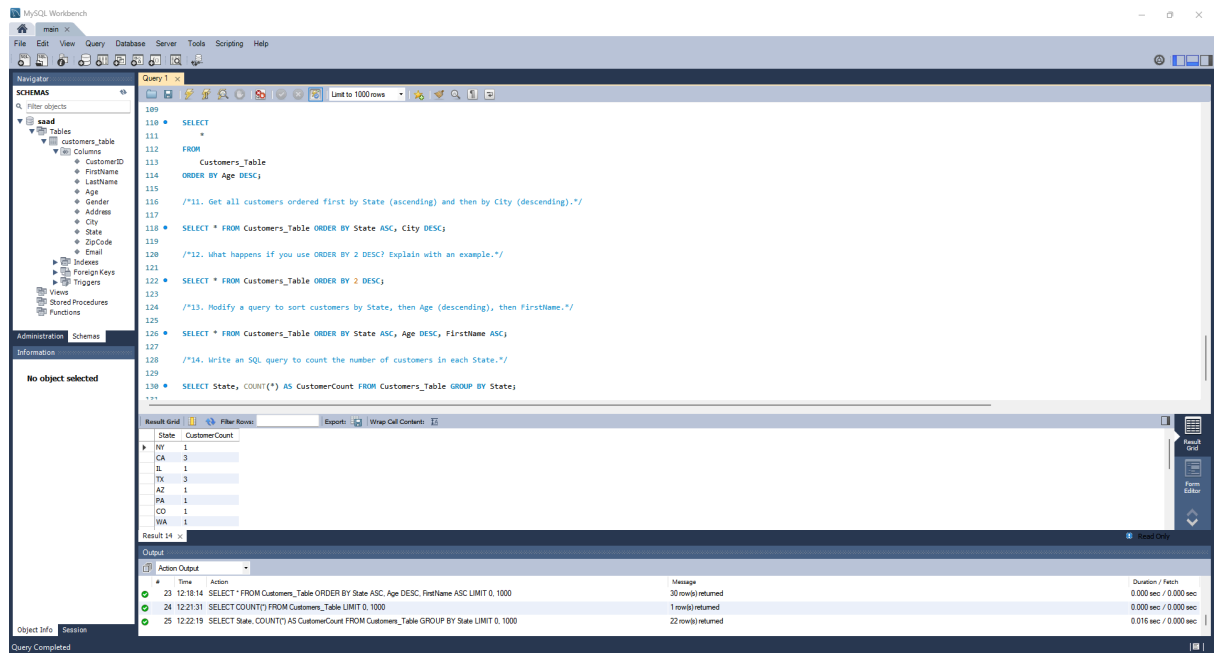
Ans. The 2 in ORDER BY 2 DESC refers to the second column in the SELECT list, which is FirstName. The result is sorted in descending order of the FirstName column. Example Screenshot attached.



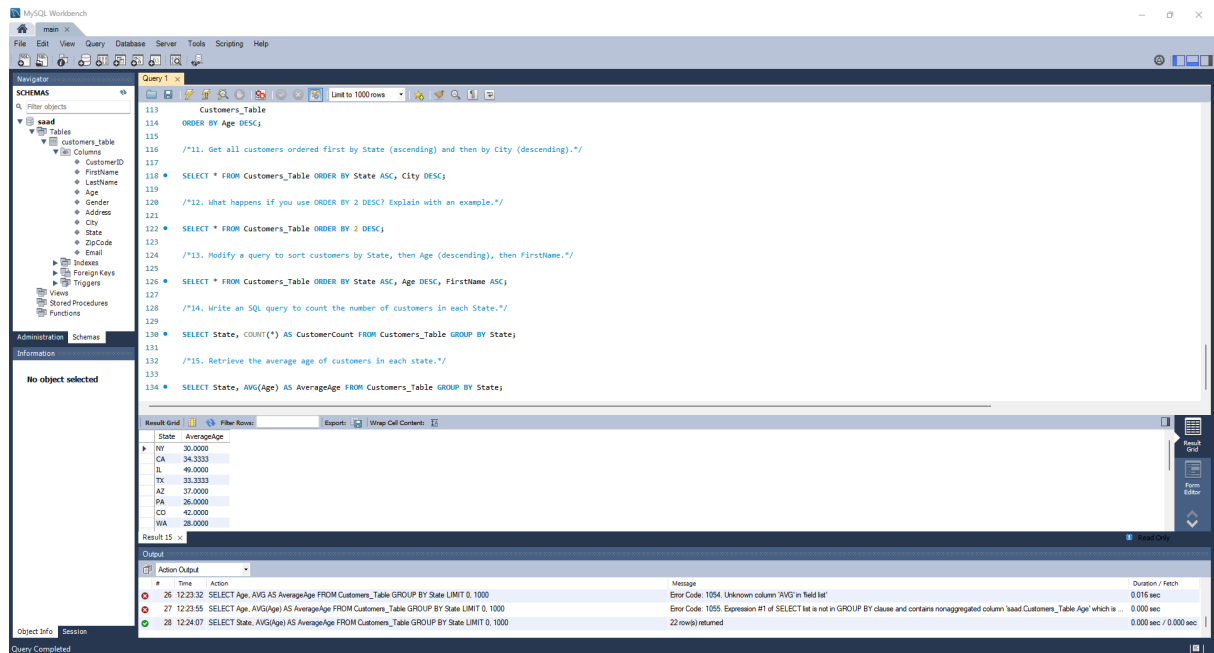
13. Modify a query to sort customers by State, then Age (descending), then FirstName.



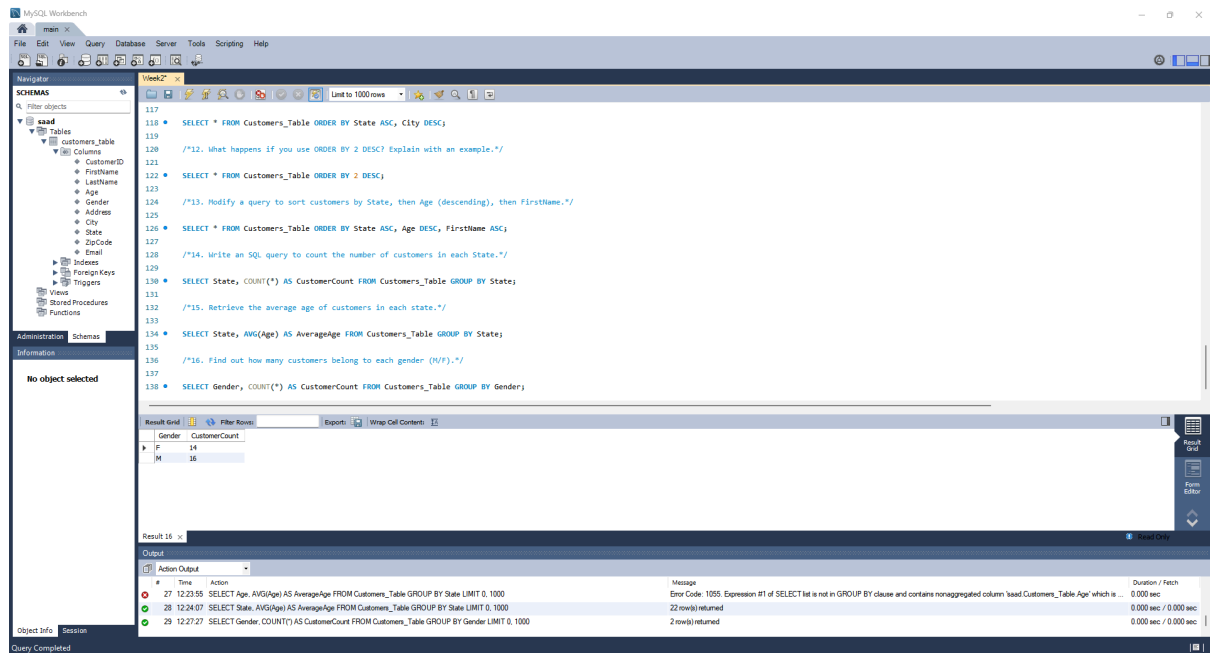
14. Write an SQL query to count the number of customers in each State.



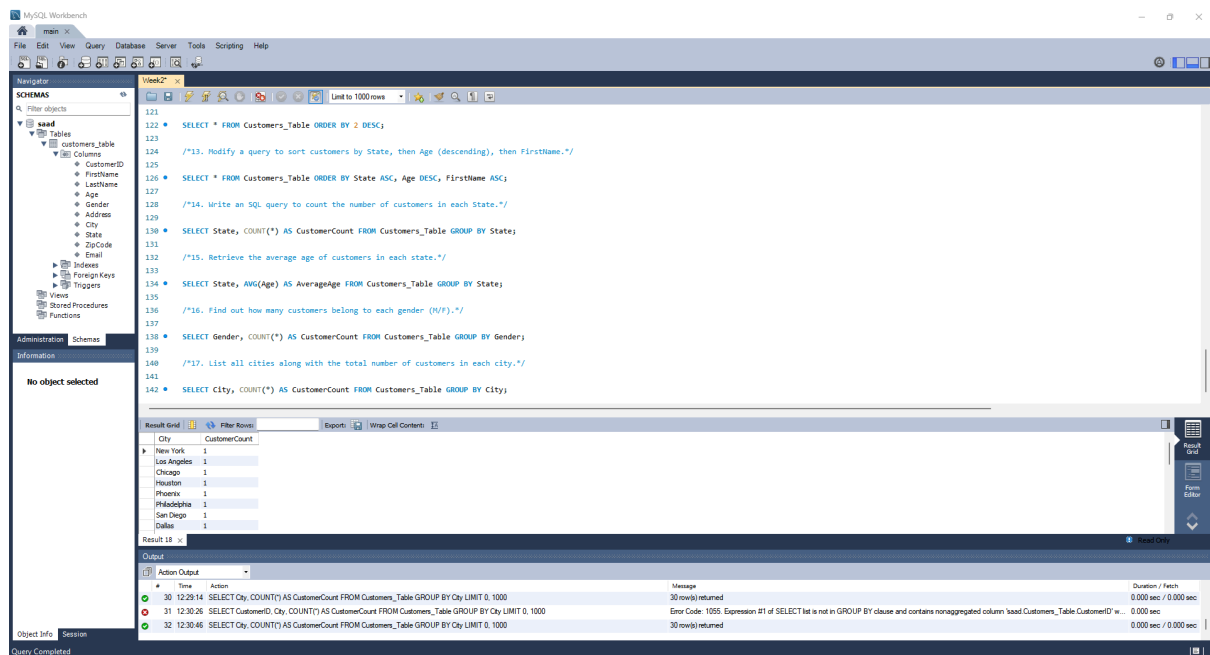
15. Retrieve the average age of customers in each state.



16. Find out how many customers belong to each gender (M/F).



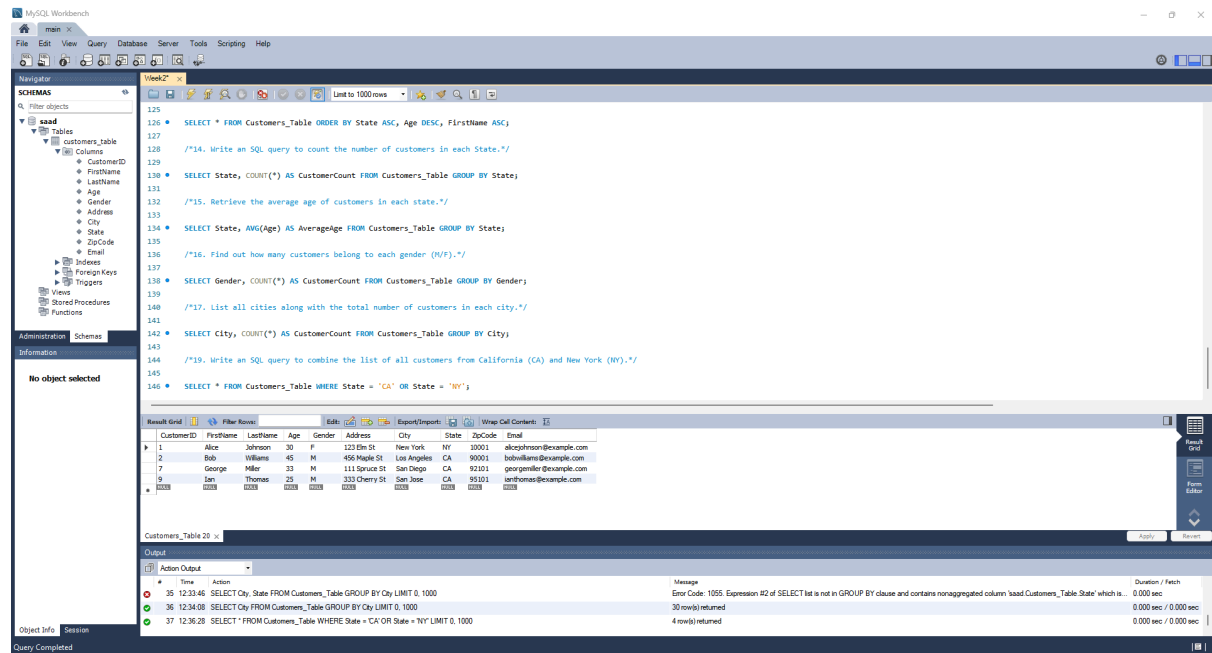
17. List all cities along with the total number of customers in each city.



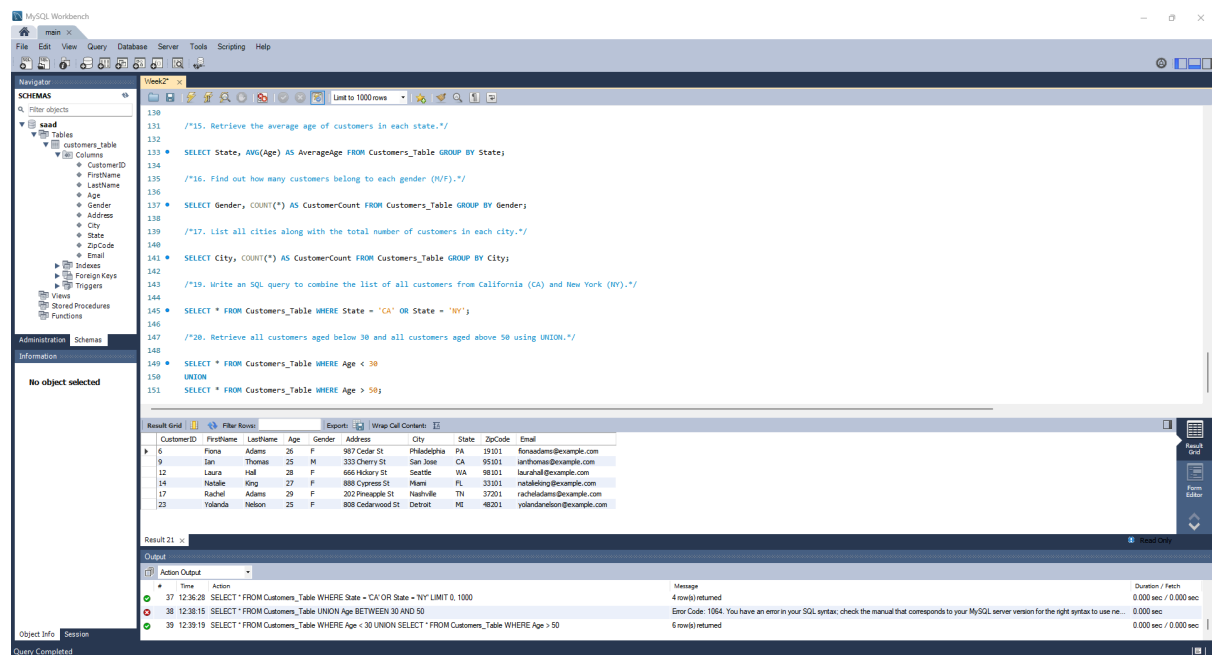
18. What will happen if you try to use GROUP BY without an aggregate function like COUNT() or SUM()?

Ans. If you use GROUP BY without an aggregate function SQL will still group the data, but the result may not be meaningful. SQL will return one row per group, but only one value for the non-grouped columns (which may be random or database-dependent).

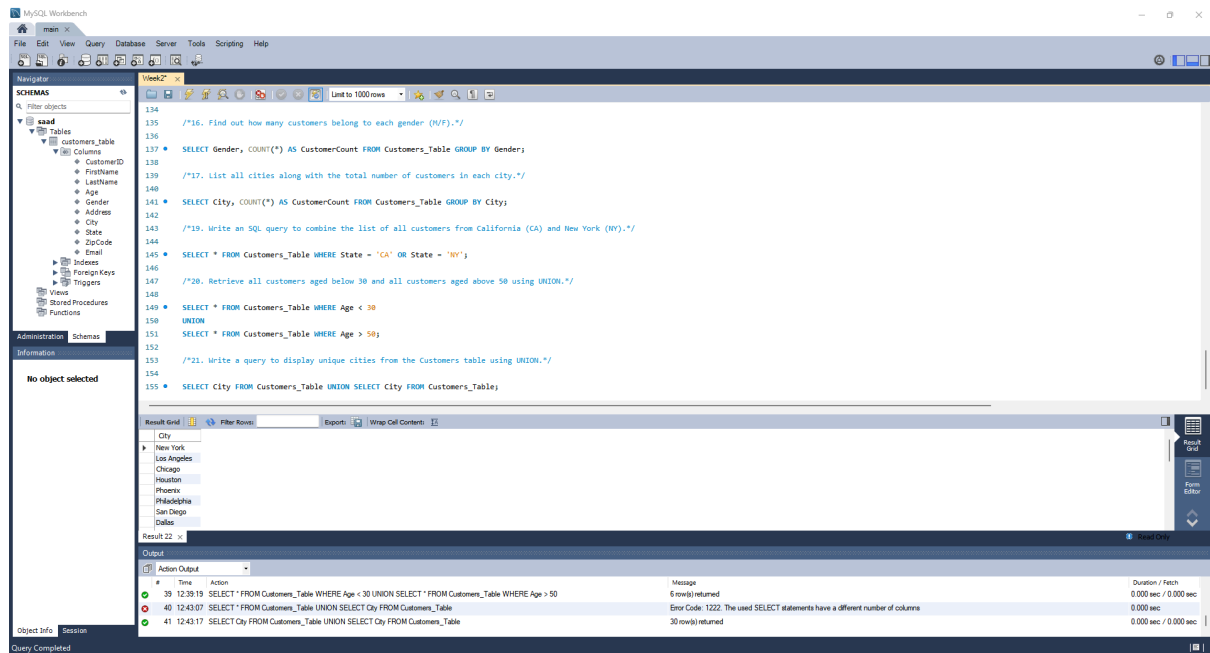
19. Write an SQL query to combine the list of all customers from California (CA) and New York (NY).



20. Retrieve all customers aged below 30 and all customers aged above 50 using UNION.



21. Write a query to display unique cities from the Customers table using UNION.



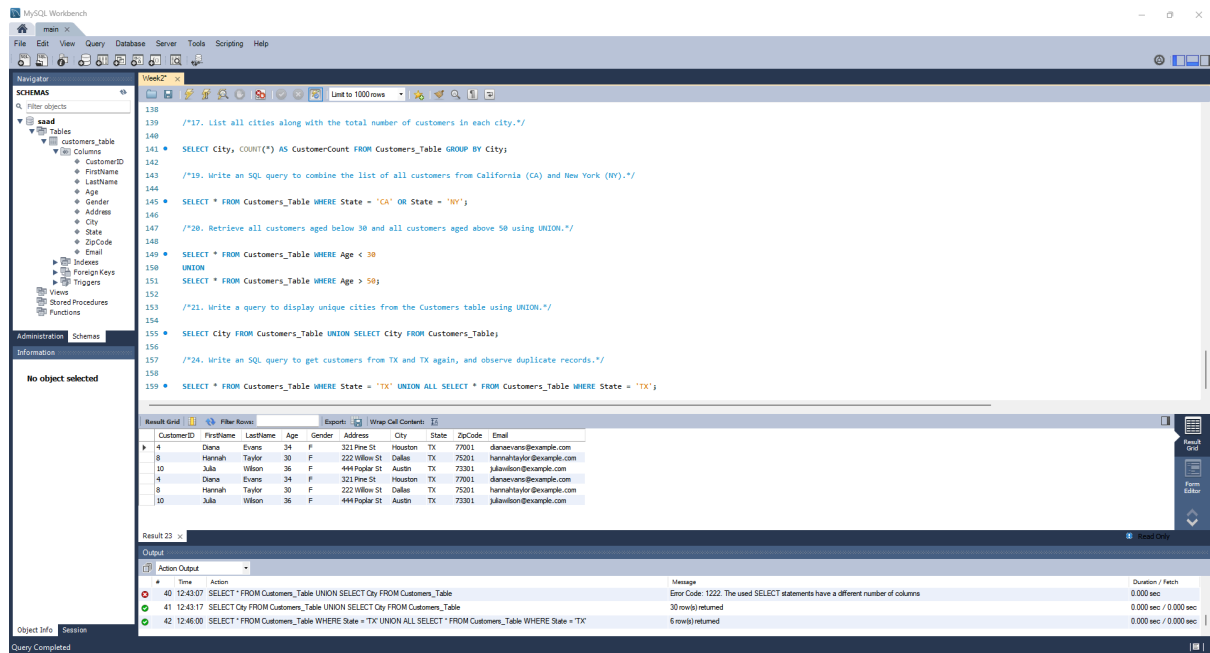
22. What happens if column counts in SELECT statements used in UNION do not match?

Ans. The query will result in an error.

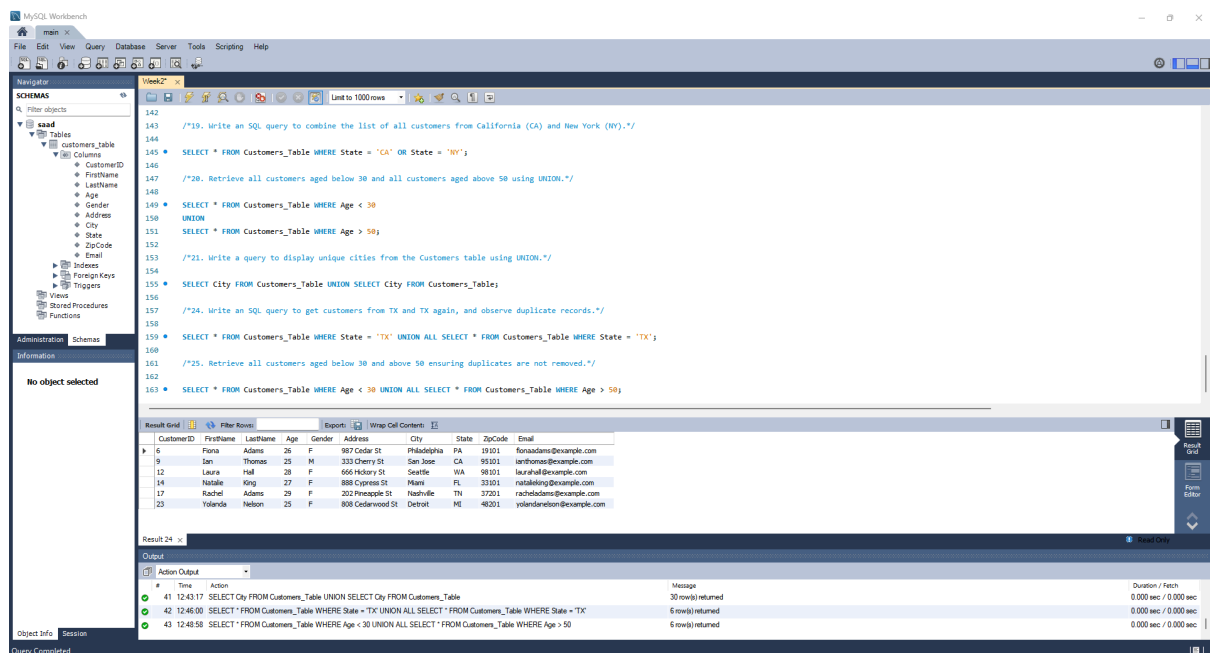
23. Why does UNION remove duplicates while UNION ALL does not? Explain with an example.

Ans. UNION removes duplicates because it performs an implicit DISTINCT operation. UNION ALL does not remove duplicates; it simply appends the results.

24. Write an SQL query to get customers from TX and TX again, and observe duplicate records.



25. Retrieve all customers aged below 30 and above 50 ensuring duplicates are not removed.



26. What happens when you use ORDER BY in a UNION ALL query? Where should it be placed?

Ans. ORDER BY must be placed at the end of the UNION ALL query and applies to the entire result set.

27. Why is UNION ALL generally faster than UNION? Explain in terms of query execution.

Ans. UNION ALL is faster than UNION because it avoids duplicate removal.

Query Execution Steps for UNION	Query Execution Steps for UNION ALL
Executes both SELECT queries.	Executes both SELECT queries.
Sorts the combined results.	Simply appends results without sorting or duplicate removal.
Removes duplicates (requires comparison of all records).	Returns the combined result set.
Returns the unique result set.	

28. What will happen if the columns in two SELECT statements used in UNION ALL have different data types?

Ans. If the columns have different data types, the database will attempt implicit type conversion. However, if the types are incompatible, it results in an error.